

Smart Metro Ticket Booking System

Project Design Phase

Date

Team ID

Project Name: Smart Metro Ticket Booking System

Maximum Marks: 4 Marks

2. Proposed Solution

The proposed solution is a ServiceNow-based Smart Metro Ticket Booking system that lets passengers look up routes, check fares, pay digitally, and get a QR ticket within minutes — while giving Station Managers, Metro Operations, and IT Admins the tools to manage stations, track performance, and configure the platform without counter dependency or custom code.

2.1 Solution Overview

At its heart, the solution digitizes the whole booking-to-travel journey. A passenger picks a source and destination from a searchable catalog form, sees an instant, itemized fare with passenger-type discounts already applied, pays via UPI, card, or wallet, and receives a unique, single-use QR ticket by app notification, email, and WhatsApp/SMS. That same QR code gets validated at the station gate in real time, with every step logged for audit, reporting, and exception handling.

2.2 End-to-End Passenger Journey

Figure 6.1 — Passenger journey from route selection to travel

2.3 Key Capabilities

- Searchable Service Catalog booking form with source/destination validation against Station Master.
- Real-time, itemized fare calculation with Student/Senior Citizen discounts applied automatically.
- Digital payments through UPI, debit/credit card, and mobile wallets, with a clear retry path on failure.
- Instant, single-use QR ticket generation linked to the passenger's booking record.
- Multi-channel delivery of tickets and alerts — in-app, email, and WhatsApp/SMS.
- Booking history and ticket re-download via the Service Portal.
- Station Manager tools for master-data upkeep and real-time gate validation.
- Metro Operations dashboards for ticket volume, revenue split, and peak-time analytics.
- IT Admin configuration of catalog variables, fare rules, and role-based access, with no code changes required.

2.4 Value Proposition

Persona	Value Delivered
Passengers	Book and travel in under 5 steps, with no queue or cash needed.
Station Managers	Faster, verifiable gate checks and less manual data upkeep.
Metro Operations	Data-driven scheduling and revenue tracking, updated in near real time.
IT Admins	Faster turnaround on fare and configuration changes, with full traceability.